

TC ID | Module | Test Case | Preconditions | Test Steps | Test Data | Expected Result | Actual Result | Status

TC001 | Registration | Verify registration with valid details

Preconditions: Application is running.

Test Steps:

1. Open Register page.
2. Enter valid name.
3. Enter valid email.
4. Enter valid password.
5. Click Register.

Test Data: Name: Smitha Email: smitha@gmail.com Password: Smith@123

Expected Result: User account should be created successfully.

Actual Result: User account created successfully.

Status: PASS

TC002 | Registration | Verify registration using existing email

Preconditions: User already exists.

Test Steps:

1. Open Register page.
2. Enter existing email.
3. Enter password.
4. Click Register.

Test Data: smitha@gmail.com

Expected Result: Application should display "Email already exists."

Actual Result: Email already exists message displayed.

Status: PASS

TC003 | Registration | Verify registration with empty fields

Preconditions: Application is running.

Test Steps:

1. Open Register page.
2. Leave all fields blank.
3. Click Register.

Test Data: Blank

Expected Result: Validation messages should appear.

Actual Result: Validation displayed.

Status: PASS

TC004 | Login | Verify login with valid credentials

Preconditions: User is registered.

Test Steps:

1. Open Login page.
2. Enter valid email.
3. Enter password.
4. Click Login.

Test Data: smitha@gmail.com Smith@123

Expected Result: Dashboard should open.

Actual Result: Dashboard opened.

Status: PASS

TC005 | Login | Verify login with invalid password

Preconditions: Registered user.

Test Steps:

1. Enter valid email.
2. Enter wrong password.
3. Click Login.

Expected Result: Invalid password error.

Actual Result: Error displayed.

Status: PASS

TC006 | Login | Verify login with empty fields

Preconditions: Application running.

Test Steps:

1. Open Login.
2. Leave fields blank.
3. Click Login.

Expected Result: Validation message displayed.

Actual Result: Validation displayed.

Status: PASS

TC007 | Logout | Verify Logout

Preconditions: User logged in.

Test Steps:

1. Click Logout.

Expected Result: Redirect to Login page.

Actual Result: Redirected successfully.

Status: PASS

TC008 | Dashboard | Verify Dashboard loads

Preconditions: User logged in.

Test Steps:

1. Login.
2. Open Dashboard.

Expected Result: Dashboard loads successfully.

Actual Result: Dashboard displayed.

Status: PASS

TC009 | Dashboard | Verify Search by Station Name

Preconditions: Dashboard loaded.

Test Steps:

1. Enter station name.
2. Observe results.

Expected Result: Matching stations displayed.

Actual Result: Matching stations displayed.

Status: PASS

TC010 | Dashboard | Verify Search by Address

Preconditions: Dashboard loaded.

Test Steps:

1. Enter address.

2. Observe results.

Expected Result: Matching stations displayed.

Actual Result: Matching stations displayed.

Status: PASS

TC011 | Dashboard | Verify Search using Contact Number

Preconditions: Dashboard is loaded.

Test Steps:

1. Enter a contact number in the search box.
2. Observe the station list.

Test Data: 9876543210

Expected Result: Matching station should be displayed.

Actual Result: Matching station displayed.

Status: PASS

TC012 | Dashboard | Verify Search with Invalid Keyword

Preconditions: Dashboard is loaded.

Test Steps:

1. Enter "XYZABC123".
2. Observe results.

Test Data: XYZABC123

Expected Result: "No Fuel Stations Found" message should be displayed.

Actual Result: No Fuel Stations Found displayed.

Status: PASS

TC013 | Dashboard | Verify Empty Search

Preconditions: Dashboard loaded.

Test Steps:

1. Keep search box empty.

Expected Result: All stations should be displayed.

Actual Result: All stations displayed.

Status: PASS

TC014 | Station Details | Verify Station Details Page Opens

Preconditions: Dashboard loaded.

Test Steps:

1. Click "View Details" for any station.

Expected Result: Station Details page opens.

Actual Result: Station Details page opened.

Status: PASS

TC015 | Station Details | Verify Fuel Availability Display

Preconditions: Station Details page opened.

Test Steps:

1. Observe Petrol, Diesel and CNG badges.

Expected Result: Correct fuel availability should be displayed.

Actual Result: Fuel availability displayed correctly.

Status: PASS

TC016 | Station Details | Verify Queue Status Display

Preconditions: Station Details page opened.

Test Steps:

1. Observe Queue Status.

Expected Result: Queue status should display Low, Medium or High.

Actual Result: Queue status displayed correctly.

Status: PASS

TC017 | Station Details | Verify Station Contact Information

Preconditions: Station Details page opened.

Test Steps:

1. Verify contact number.

Expected Result: Correct contact number displayed.

Actual Result: Correct contact displayed.

Status: PASS

TC018 | Map | Verify User Location Detection

Preconditions: Location permission enabled.

Test Steps:

1. Open Map page.
2. Allow browser location permission.

Expected Result: Current user location should be displayed.

Actual Result: Current location displayed successfully.

Status: PASS

TC019 | Map | Verify Nearest Fuel Stations

Preconditions: Location detected.

Test Steps:

1. Open Map page.

Expected Result: Stations should be sorted by nearest distance.

Actual Result: Nearest stations displayed correctly.

Status: PASS

TC020 | Map | Verify Google Maps Navigation

Preconditions: Map page loaded.

Test Steps:

1. Click "Open in Google Maps".

Expected Result: Google Maps should open with selected station.

Actual Result: Google Maps opened successfully.

Status: PASS

TC021 | Profile | Verify Profile Page Loads

Preconditions: User is logged in.

Test Steps:

1. Click Profile from the navigation bar.

Expected Result: Profile page should open successfully.

Actual Result: Profile page opened successfully.

Status: PASS

TC022 | Profile | Verify User Information Display

Preconditions: Profile page is opened.

Test Steps:

1. Observe user details such as Name and Email.

Expected Result: Correct user information should be displayed.

Actual Result: Correct information displayed.

Status: PASS

TC023 | Admin | Verify Admin Dashboard Loads

Preconditions: Admin user is logged in.

Test Steps:

1. Open Admin Dashboard.

Expected Result: Admin Dashboard should load successfully.

Actual Result: Dashboard loaded successfully.

Status: PASS

TC024 | Admin | Verify Statistics Cards Display

Preconditions: Admin Dashboard opened.

Test Steps:

1. Observe dashboard statistics.

Expected Result: Total Stations, Open Stations, Closed Stations and other statistics should be displayed.

Actual Result: Statistics displayed correctly.

Status: PASS

TC025 | Admin | Verify Add New Fuel Station

Preconditions: Admin Dashboard opened.

Test Steps:

1. Click Add Station.
2. Enter valid station information.
3. Click Save.

Test Data: Station Name: HP Panvel Status: Open

Expected Result: New station should be added successfully.

Actual Result: Station added successfully.

Status: PASS

TC026 | Admin | Verify Edit Fuel Station

Preconditions: Station already exists.

Test Steps:

1. Click Edit.
2. Modify station information.
3. Save changes.

Expected Result: Station information should be updated.

Actual Result: Station updated successfully.

Status: PASS

TC027 | Admin | Verify Delete Fuel Station

Preconditions: Station exists.

Test Steps:

1. Click Delete.
2. Confirm deletion.

Expected Result: Selected station should be deleted.

Actual Result: Station deleted successfully.

Status: PASS

TC028 | Admin | Verify CSV Export

Preconditions: Admin Dashboard loaded.

Test Steps:

1. Click Export CSV.

Expected Result: CSV file should be downloaded successfully.

Actual Result: CSV downloaded successfully.

Status: PASS

TC029 | Admin | Verify Newly Added Station Appears on Dashboard

Preconditions: Admin has added a new station.

Test Steps:

1. Open Dashboard.

Expected Result: Newly added station should be visible.

Actual Result: Station displayed successfully.

Status: PASS

TC030 | Admin | Verify Updated Station Details Reflect on Dashboard

Preconditions: Admin has edited a station.

Test Steps:

1. Open Dashboard.
2. Search the updated station.

Expected Result: Updated information should be displayed.

Actual Result: Updated information displayed successfully.

Status: PASS

TC031 | Dashboard | Verify Search is Case Insensitive

Preconditions: Dashboard is loaded.

Test Steps:

1. Enter station name in lowercase.
2. Enter the same station name in uppercase.

Expected Result: Search should return the same matching station regardless of letter case.

Actual Result: Search works correctly for different letter cases.

Status: PASS

TC032 | Dashboard | Verify Search with Special Characters

Preconditions: Dashboard is loaded.

Test Steps:

1. Enter @#\$%^&* in the search box.

Expected Result: No application crash should occur. "No Fuel Stations Found" message should be displayed.

Actual Result: Application handled invalid input correctly.

Status: PASS

TC033 | Dashboard | Verify Station Card Information

Preconditions: Dashboard loaded.

Test Steps:

1. Observe each station card.

Expected Result: Station Name, Address, Status, Queue, Contact, Fuel Availability and Last Updated should be displayed.

Actual Result: All information displayed correctly.

Status: PASS

TC034 | Map | Verify Location Permission Allowed

Preconditions: Browser asks for location permission.

Test Steps:

1. Click Allow.

Expected Result: Current location should be displayed.

Actual Result: Current location displayed successfully.

Status: PASS

TC035 | Map | Verify Location Permission Denied

Preconditions: Browser asks for location permission.

Test Steps:

1. Click Block.

Expected Result: Application should display an appropriate error message.

Actual Result: Location permission denied message displayed.

Status: PASS

TC036 | Map | Verify Distance Calculation

Preconditions: Location detected.

Test Steps:

1. Open Map page.
2. Observe calculated distance.

Expected Result: Distance should be displayed in kilometers.

Actual Result: Distance displayed correctly.

Status: PASS

TC037 | Navigation | Verify Navbar Links

Preconditions: User logged in.

Test Steps:

1. Click Dashboard.
2. Click Map.
3. Click Profile.
4. Click Admin.

Expected Result: Each page should open correctly.

Actual Result: Navigation works correctly.

Status: PASS

TC038 | Navigation | Verify Footer Display

Preconditions: Any page opened.

Test Steps:

1. Scroll to the bottom.

Expected Result: Footer should be displayed properly.

Actual Result: Footer displayed correctly.

Status: PASS

TC039 | Responsive UI | Verify Mobile View

Preconditions: Application opened.

Test Steps:

1. Open browser Developer Tools.
2. Switch to Mobile View.

Expected Result: Application should adjust without layout issues.

Actual Result: Responsive layout works correctly.

Status: PASS

TC040 | Error Handling | Verify Invalid URL

Preconditions: Application running.

Test Steps:

1. Enter an invalid URL such as /abcxyz.

Expected Result: 404 Not Found page should be displayed.

Actual Result: Not Found page displayed successfully.

Status: PASS

TC041 | Login | Verify Login with Invalid Email Format

Preconditions: Application is running.

Test Steps:

1. Open Login page.
2. Enter an invalid email format.
3. Enter any password.
4. Click Login.

Test Data: abc123

Expected Result: Application should display "Invalid Email Format".

Actual Result: Validation message displayed.

Status: PASS

TC042 | Registration | Verify Password Length Validation

Preconditions: Application is running.

Test Steps:

1. Open Registration page.
2. Enter valid name and email.
3. Enter a password with less than 6 characters.
4. Click Register.

Expected Result: Password validation message should be displayed.

Actual Result: Validation displayed successfully.

Status: PASS

TC043 | Dashboard | Verify Station List After Login

Preconditions: User logged in.

Test Steps:

1. Login.

2. Open Dashboard.

Expected Result: Fuel station list should load successfully.

Actual Result: Station list displayed.

Status: PASS

TC044 | Profile | Verify User Email Display

Preconditions: User logged in.

Test Steps:

1. Open Profile page.

Expected Result: Logged-in user's email should be displayed.

Actual Result: Correct email displayed.

Status: PASS

TC045 | Security | Verify Unauthorized Access to Admin Page

Preconditions: Normal user logged in.

Test Steps:

1. Try opening Admin page directly.

Expected Result: Access should be denied or redirected.

Actual Result: Unauthorized access prevented.

Status: PASS

TC046 | Session | Verify Session Persistence

Preconditions: User logged in.

Test Steps:

1. Login.
2. Refresh browser.

Expected Result: User should remain logged in.

Actual Result: Session maintained successfully.

Status: PASS

TC047 | Browser Compatibility | Verify Application in Google Chrome

Preconditions: Application deployed.

Test Steps:

1. Open application in Google Chrome.

Expected Result: All features should work correctly.

Actual Result: Application works correctly.

Status: PASS

TC048 | Performance | Verify Dashboard Loading Time

Preconditions: Application running.

Test Steps:

1. Login.
2. Open Dashboard.

Expected Result: Dashboard should load within an acceptable time.

Actual Result: Dashboard loaded successfully.

Status: PASS

TC049 | Error Handling | Verify Internet Connection Loss

Preconditions: Application running.

Test Steps:

1. Disconnect internet.
2. Refresh application.

Expected Result: Appropriate error message should be displayed.

Actual Result: Application handled network error appropriately.

Status: PASS

TC050 | Logout | Verify Session Ends After Logout

Preconditions: User logged in.

Test Steps:

1. Click Logout.
2. Press browser Back button.

Expected Result: Dashboard should not be accessible without logging in again.

Actual Result: User redirected to Login page.

Status: PASS